

Analytical Modeling of a Membrane-Based Pneumatic Soft Gripper

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Abstract—Finite deformation is the principal actuation basis of elastomer-based pneumatic soft actuators. Desired deformation behavior is the key design requirement for such actuators. The objective of current study is to optimize the design of a flat shell gripper and to investigate its interaction with a cylindrical object. Herein, we propose an analytical model for a membrane-based flat shell gripper. The model is based on finite strain membrane theory and neo-Hookean material. The proposed model considers the contact interaction of the actuator with flat and cylindrical rigid substrates. The model is developed for three different states of the actuator: 1) free-space; 2) contact with a flat substrate; and 3) contact with a cylindrical substrate. In application, the model was used to predict the relative position and air pressure required to grasp a cylindrical object by a parallel two-fingered shell gripper. Additionally, the frictional behavior of the actuator in contact with a cylindrical substrate is investigated. The model involves only solving nonlinear algebraic equations and is computationally efficient. The theoretically predicted deformation behavior of the actuator is experimentally validated via free-space deformation, force measurement, and grasping tests.

Index Terms—Soft actuator, shell gripper, analytical modeling, contact force, grasping.

I. INTRODUCTION

SOFT actuators have the potential to serve as a versatile gripper for grasping variety of objects with moderate precision. Pneumatic soft grippers made of elastomers have recently gained extensive attention among researchers because of their low fabrication cost, maintenance cost, and wide application domain [1]. Recent advancement in 3D printing technology also enables the fabrication of these actuators as a single entity. These grippers are best suited for grasping objects with

highly irregular geometry, fragile properties, and objects in an unknown environment [2]. When actuated, elastomeric soft actuators deformed in a designed geometric shape [3]. However, the design of these actuators is challenging because of their large deformation behavior. Many of these actuators have a membrane-based design [4]–[9]. Membrane-based structures generate large deformations even at low actuation pressures and hence are well suited for such actuators. Choi *et al.* [10] conducted a study on the design and feasibility of a flexible gripper made of inflatable rubber pockets. Wang *et al.* [5], [11] presented circular and parallel shell grippers for handling food products. A membrane-based needle gripper for grasping and piercing food materials was presented in [12]. Based on soft membrane, a bistable valve for the autonomous control of a soft actuator was presented in [13]. A soft-bubble gripper for robust and perceptive manipulation was presented in [14]. Membrane-based jamming and self-sealing grippers were presented in [15], [16].

To understand the deformation behavior, the modeling of such actuators, which involves both geometric and material nonlinearities, has attracted significant attention [17]–[19]. Finite element (FE)-based simulations are widely used for the structural optimization of these actuators [20]–[23]. Coevoet *et al.* [22] presented a FE-based unified software framework for the modeling, simulation, and control of soft robots. Additionally, analytical models for soft actuators have been developed [24]–[27]. Finite strain models for membrane-based soft actuator were presented in [4], [28], [29]. To model a pneumatically driven membrane with clamped edge conditions, various finite-strain-based modeling approaches have been presented [30], [31].

The analytical modeling of these actuators is in its developmental phase and is not sufficiently mature. This study focuses on the contact interaction between a flat shell gripper and a cylindrical object. Based on finite strain membrane theory and incompressible neo-Hookean material, an analytical model is proposed for the actuator. The model was developed for three different states of the actuator: 1) free-space; 2) contact with a flat substrate; and 3) contact with a cylindrical substrate. The key contributions of current study are as follows:

- The proposed model predicts the force characteristics of the actuator for both flat and cylindrical contacts.
- The proposed model can be used in grasping to predict the air pressure required to attain a grasp.
- Based on deformed profile of the actuator, the proposed model also predicts the desired relative position of the actuator to attain a grasp (i.e., offset h).

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- The friction between the actuator and cylindrical object is investigated.
- The proposed model is computationally efficient and requires less computational resources.

In application, the proposed model was used to predict the air pressure and relative position of the actuator for grasping cylindrical objects; furthermore, it was validated experimentally.

II. THEORETICAL FORMULATION

A. Actuator Design and Theoretical Assumptions

The actuator comprised a rigid shell and a soft thin membrane. The rigid shell, which is typically fabricated using a 3D printer with PLA or PETG materials, serves not only the mold for casting the thin membrane, but also as a component of the actuator for providing support to the membrane. The fabrication method is relatively simple and it requires only a one-time casting procedure that does not involve demolding. The detailed design and fabrication of a similar actuator was presented in [11]. The actuator is pneumatically driven and can generate a relatively large force because of the rigid shell. Certain assumptions were made on the design and behavior of the actuator to reduce modeling complexity. The actuator is assumed to be a rectangular membrane with clamped end condition. The pressure is assumed to be uniformly distributed on the membrane and at the contact region upon grasping an object.

B. Membrane Theory

For hyperelastic membranes, the stress resultants or line tensions T_i in the principal directions are defined as [30], [32]

$$\begin{aligned} T_1 &= 2t_m \lambda_3 (\lambda_1^2 - \lambda_3^2) (U_1 + \lambda_2^2 U_2) \\ T_2 &= 2t_m \lambda_3 (\lambda_2^2 - \lambda_3^2) (U_1 + \lambda_1^2 U_2) \end{aligned} \quad (1)$$

where U_i denotes $\partial U / \partial I_i$. The equilibrium equation for a membrane in terms of curvature κ_i and line tensions is defined as

$$\kappa_1 T_1 + \kappa_2 T_2 = p \quad (2)$$

where p is the lateral pressure. For a rectangular membrane, the change in dimensions along the principal directions is assumed to be $\Delta a = \Delta b$ [see Fig. 1(a)]. Using this assumption, the principal stretches λ_1 and λ_2 can be expressed as

$$\lambda_1 = \lambda; \text{ and } \lambda_2 = 1 + \frac{a}{b}(\lambda - 1). \quad (3)$$

The membrane material is assumed to be isotropic and incompressible. For an incompressible material, the principal stretch λ_3 can then be written as

$$\lambda_3 = (\lambda_1 \lambda_2)^{-1}. \quad (4)$$

The strain energy density function U for a neo-Hookean material [33] is defined by

$$U = \frac{\mu}{2} (I_1 - 3). \quad (5)$$

After substituting (5) into (1), the line tensions become

$$T_1 = \mu t_m \lambda_3 (\lambda_1^2 - \lambda_3^2)$$

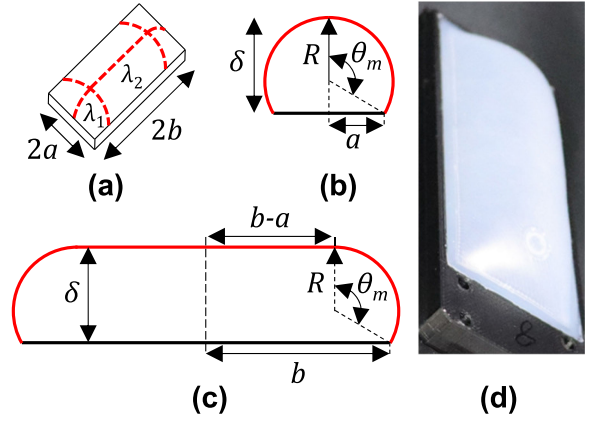


Fig. 1. Schematic illustration of free deformation of a rectangular membrane. (a) Isometric view, (b) section view along actuator width, (c) section view along actuator length, and (d) experimental snapshot of inflated actuator.

$$T_2 = \mu t_m \lambda_3 (\lambda_2^2 - \lambda_3^2). \quad (6)$$

Now, the only undefined parameter in (2) is curvature κ_i . Next, based on the deformed geometry, we define the curvature for all three states of the actuator.

C. Free-Space Deformation

The deformed profile along actuator width is assumed to exhibit a circular shape with curvature $\kappa_1 = 1/R$ [see Fig. 1(b)]. The average curvature [34] along actuator length can be written as [due to symmetry, only one-half is considered; see Fig. 1(c)]

$$\kappa_2 = \frac{\theta_m}{R \theta_m + (b - a)}. \quad (7)$$

Based on deformed geometry of the membrane, we can express the radius of curvature R as

$$R = \frac{a}{\sin \theta_m}. \quad (8)$$

The stretch λ_1 along actuator width can be written as

$$\lambda_1 = \frac{\theta_m}{\sin \theta_m}. \quad (9)$$

The maximum inflation δ of the membrane is derived from the deformed geometry as

$$\delta = a \tan \left(\frac{\theta_m}{2} \right). \quad (10)$$

After substituting the curvatures into (2), it can be written as

$$\frac{1}{R} T_1 + \frac{\theta_m}{R \theta_m + (b - a)} T_2 = p. \quad (11)$$

Equation (11) is the final governing equation for free-space deformation of the membrane. After substituting (3), (4), (6), (8), and (9) into (11), it reduced to a nonlinear algebraic equation, having θ_m as the only variable.

TABLE I
LIST OF SYMBOLS

Symbol	Description	Symbol	Description
$2a$	Width of the membrane	μ_f	Coefficient of friction
$2b$	Length of the membrane	$2c_a$	Contact width along actuator width
t_m	Thickness of the membrane	$2c_b$	Contact width along actuator length
T_i	Stress resultant or line tension	A_c	Contact area
U	Strain energy density function	d	Object offset (horizontal)
I_i	Strain invariant	h	Object offset (vertical)
λ_i	Stretch along principal directions	e_i	Deformed portion of the membrane with zero curvature
κ_i	Curvature	r	Radius of cylindrical object
R	Radius of curvature	μ	Material shear modulus
R_a	Radius of curvature (cylindrical substrate; section z_0 – z_0)	δ	Membrane mid-plane extension
R_c	Radius of curvature at contact edge (section z_1 – z_1)	p	Air pressure
θ_m	Arc angle	F_c	Contact force
θ_{m_a}	Arc angle (cylindrical substrate; section z_0 – z_0)	F_{ext}	Required normal force for grasping
θ_{m_c}	Arc angle at contact edge (section z_1 – z_1)	F_L	Lifting force
θ_c	Contact angle	F_z	Component of F_c in Z-direction
λ_{10}	Stretch along actuator width at contact edge (section z_1 – z_1)	mg	Weight of grasping object

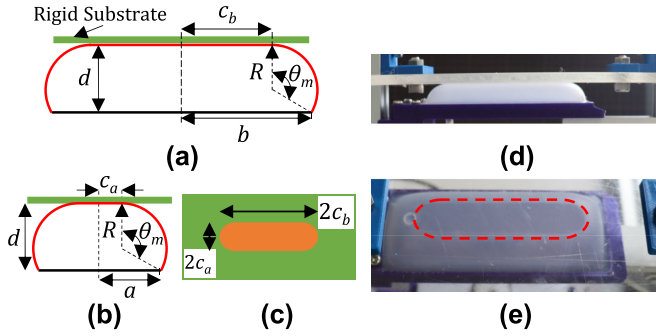


Fig. 2. Schematic illustration of a rectangular membrane in contact with a flat rigid substrate. (a) Section view along actuator length, (b) section view along actuator width, (c) assumed shape of contact region, (d) experimental setup to determine actual shape of contact region, and (e) actual contact region. In (e), dotted red curve represents an oval shape.

D. Membrane in Contact With Flat Substrate

From the deformed geometry [see Fig. 2(a)], the radius of curvature R can be written as

$$R = \frac{d}{1 - \cos \theta_m}. \quad (12)$$

The contact width c_a along actuator width [see Fig. 2(b)] can be written as

$$c_a = a - d \cot \left(\frac{\theta_m}{2} \right). \quad (13)$$

The contact width c_b along actuator length can be expressed in terms of c_a as

$$c_b = \lambda_2 b - \lambda_1 a + c_a. \quad (14)$$

The stretch λ_1 along actuator width can be expressed as

$$\lambda_1 = 1 + \frac{d}{a} \left[\frac{\theta_m}{2} \csc^2 \left(\frac{\theta_m}{2} \right) - \cot \left(\frac{\theta_m}{2} \right) \right]. \quad (15)$$

After substituting (3), (4), (6), (12), and (15) into equilibrium (11), it get reduced to a nonlinear algebraic equation, having θ_m as the only variable. The contact region exhibits an oval shape, as validated experimentally in Fig. 2(d) and (e). The contact

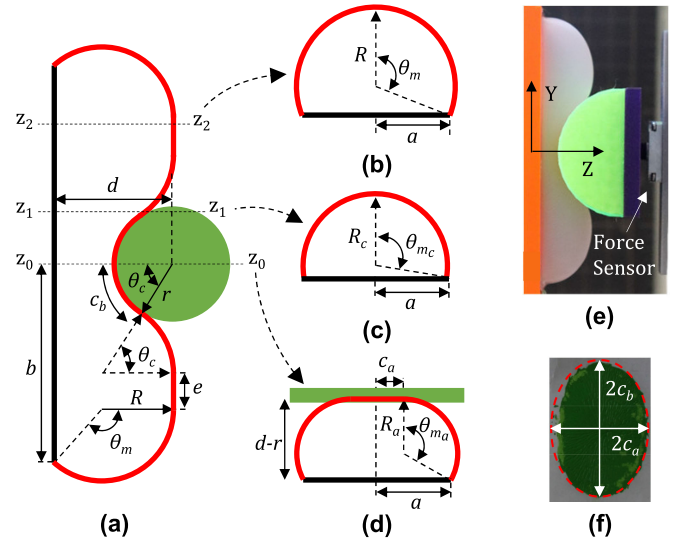


Fig. 3. Schematic illustration of a rectangular membrane in contact with a rigid cylindrical substrate. (a) Section view of contact along actuator length, (b) free deformation along actuator width (section z_2 – z_2), (c) deformation at contact edge (section z_1 – z_1), (d) contact at the mid-plane (section z_0 – z_0), (e) photo snapshot of experimental setup for force measurement, and (f) experimentally validated assumed shape of elliptical contact region.

area A_c for an oval-shaped region [see Fig. 2(c)] can then be calculated as

$$A_c = 2c_a (2c_b - 2c_a) + \pi c_a^2. \quad (16)$$

The contact force F_c can be calculated as

$$F_c = p A_c. \quad (17)$$

III. GRASPING CYLINDRICAL-SHAPED OBJECT

In this section, we formulated the model of membrane actuator in contact with a cylindrical substrate for applications in grasping agricultural products, such as cucumbers. Here, we consider the axis of substrate to be parallel to the width of actuator, as shown in Fig. 3.

A. Contact Along Actuator Length $2b$

Based on deformed geometry of the membrane [see Fig. 3(a)], stretch λ_2 can be expressed as (due to symmetry, only one-half is considered)

$$\lambda_2 = \frac{r \theta_c + R \theta_c + e + R \theta_m}{b}. \quad (18)$$

The undeformed length b can be expressed as

$$b = r \sin \theta_c + R \sin \theta_c + e + R \sin \theta_m. \quad (19)$$

The object offset d can be expressed as

$$d = r \cos \theta_c + R \cos \theta_c - R \cos \theta_m. \quad (20)$$

The contact width c_b along actuator length can be written as

$$c_b = r \theta_c. \quad (21)$$

B. Contact Along Actuator Width $2a$

The deformed profile of the membrane is assumed to be spherical at the ends [see Fig. 1(a)]. From the deformed geometry for section $z_2 - z_2$, as shown in Fig. 3(b), the undeformed width a can be expressed as

$$a = R \sin \theta_m. \quad (22)$$

The stretch λ_1 along actuator width can be expressed as

$$\lambda_1 a = R \theta_m. \quad (23)$$

From the deformed geometry for section $z_0 - z_0$ as shown in Fig. 3(d), the object offset can be expressed as

$$d - r = R_a - R_a \cos \theta_{m_a}. \quad (24)$$

The undeformed width a of the membrane can be expressed as

$$a = c_a + R_a \sin \theta_{m_a}. \quad (25)$$

Along actuator width, the stretch is assumed to be constant throughout the contact region, i.e., the stretch at section $z_0 - z_0$ and $z_1 - z_1$ are equal [see Fig. 3(c) and (d)]. From deformed geometry, the stretch λ_{1_0} at section $z_1 - z_1$ can be calculated as

$$\lambda_{1_0} = \frac{\theta_{m_c}}{\sin \theta_{m_c}} = \frac{2 \tan^{-1} \left[\frac{R (\cos \theta_c - \cos \theta_m)}{a} \right]}{\sin \left\{ 2 \tan^{-1} \left[\frac{R (\cos \theta_c - \cos \theta_m)}{a} \right] \right\}}. \quad (26)$$

The stretch λ_{1_0} can be expressed as

$$\lambda_{1_0} a = c_a + R_a \theta_{m_a}. \quad (27)$$

The contact region is assumed to be an ellipse and is validated experimentally, as shown in Fig. 3(f). For ellipse, the contact area can be calculated as

$$A_c = \pi c_a c_b. \quad (28)$$

In the force-measuring experiment, the sensor recorded force data corresponding to the Z-direction, as shown in Fig. 3(e). The component F_z of the contact force along the Z-direction

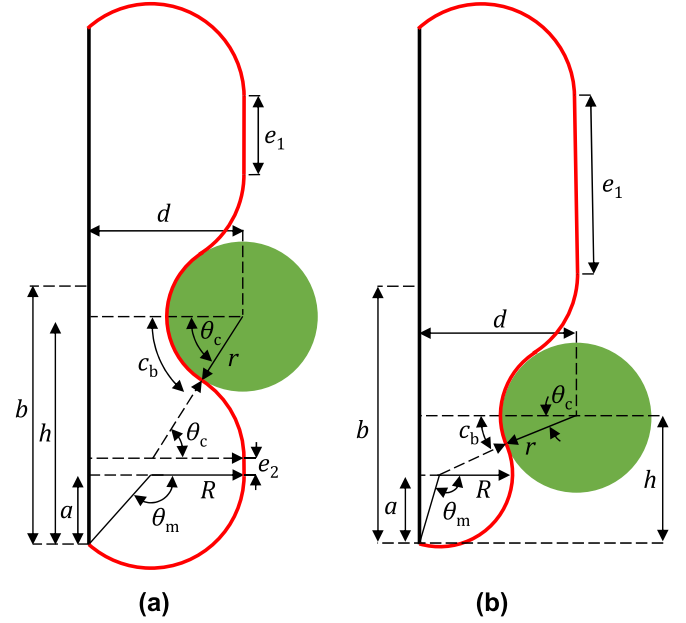


Fig. 4. Schematic illustration of a rectangular membrane in contact with a rigid cylindrical substrate. (a) Offset configuration ($e_2 > 0$) and (b) offset configuration $h < (r + a)$ or $e_2 < 0$.

can be calculated by projecting the contact area normal to the Z-direction as

$$F_z = p \pi c_a r \sin \theta_c. \quad (29)$$

C. Curvature

Along actuator width, the stretch is maximum at section $z_2 - z_2$. The curvature at this section is: $\kappa_1 = 1/R$. The average curvature of free portion of the membrane along actuator length is defined as

$$\kappa_2 = \frac{\theta_m + \theta_c}{R \theta_c + e + R \theta_m}. \quad (30)$$

We now have sufficient number of equations for the number of unknown variables. Therefore, by solving (2), (4), (6), (18), (19), (20), (22), (23), (24), (25), (26), (27), and (30) as a set of nonlinear algebraic equations, we can obtain all the unknown variables.

D. Offset Configuration

The vertical offset between the membrane and cylindrical object is defined by $b - h$, as shown in Fig. 4(a). Based on the assumed deformed profile of the membrane, a limit exists on h , below which the membrane cannot grasp the object. The restriction on h is possessed by the parameter e_2 . A successful grasp can only be achieved if $e_2 \geq 0$. For $e_2 < 0$, the object does not get a stable enclosure because the net resultant grasping force is in the downward direction [see Fig. 4(b)]. For $e_2 > 0$, the contact along actuator width remains the same as that for $b = h$. Here, the deformed profile along actuator length becomes asymmetric, as shown in Fig. 4(a). For this case, the stretch λ_2

can be defined as

$$\lambda_2 = \frac{2r \theta_c + 2R \theta_c + e_1 + 2R \theta_m + e_2}{2b}. \quad (31)$$

The undeformed length b can be expressed as

$$2b - h = r \sin \theta_c + R \sin \theta_c + e_1 + R \sin \theta_m. \quad (32)$$

The vertical offset h can be expressed as

$$h = r \sin \theta_c + R \sin \theta_c + e_2 + R \sin \theta_m. \quad (33)$$

The horizontal object offset d can be expressed as

$$d = r \cos \theta_c + R \cos \theta_c - R \cos \theta_m. \quad (34)$$

For such configurations, the contact along actuator width and the remaining equations are the same as those for $b = h$.

E. Offset Height h for Stable Grasp

The minimum value of offset height h for a successful grasp can be calculated corresponds to $e_2 = 0$. We can calculate h by substituting $e_2 = 0$ into (31) and (33) and then solving the system of nonlinear equations for h . Vertical offset is a critical parameter for achieving a grasp by the gripper.

IV. SIMULATIONS AND EXPERIMENTS

A. Simulations

The governing equations for the actuator are nonlinear algebraic equations. The equations were solved numerically using nonlinear algebraic equation solver “fsolve, MATLAB R2019b”. The computations were performed on a desktop computer [Dell Optiplex; Windows 10 Pro; Intel Core i7-9700 CPU @3.00 GHz ×64-based processor; 8 GB RAM]. The average computation time was 1.57 s. The numerical solution of a nonlinear equation depends on the initial guess. In the simulations, the initial guesses were chosen on the basis of practically feasible solution range. The simulations were performed for four different states of the actuator: 1) free-space; 2) contact with a flat substrate; 3) contact with a cylindrical substrate; and 4) grasping cylindrical objects. The pseudocode of our solution algorithm for the grasping state is presented in Algorithm 1. Here, the pressure value for the initial guess is the pressure required for contact initiation. The initial guess pressure can be calculated from the equations for free-space state by substituting $\delta = d$ into (10) and then solving the system of equation for p . The cut-off condition or the acceptable force error ϵ_f was set to 10^{-6} N. The geometric dimensions of the membrane are: $2a = 32$ mm; $2b = 86$ mm; and $t_m = 1.1$ mm.

B. Experiments

Experiments were performed for free-space deformation, force measurement, and grasping. Air pressure was generated using an off-board compressor (ANEST IWATA SLP-15EFD) and controlled using a digital pressure regulator (SMC ITV2030). The deformation of the actuators was captured using a camera (Canon Inc.). The camera images were processed for lens distortion using “MATLAB R2019b Single Camera

Algorithm 1 Grasping Cylindrical Objects

Input: F_{ext} , a , b , d , r , t_m , μ

Output: θ_c , θ_m , R , θ_m , e_1 , h , p

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1  $p \leftarrow p_i$  {initial guess pressure}
2 repeat
3  $\theta_c, \theta_m, R, \theta_m, e_1, h \leftarrow$  Solve (2), (22), (27), (32),
   (33), and (34) as a set of nonlinear algebraic equations
4  $F_c \leftarrow$  Solve (17) and (28)
5  $p \leftarrow p + \Delta p$ 
6 until  $|F_{ext} - F_c| \leq \epsilon_f$  or  $(F_{ext} - F_c) \leq 0$ 
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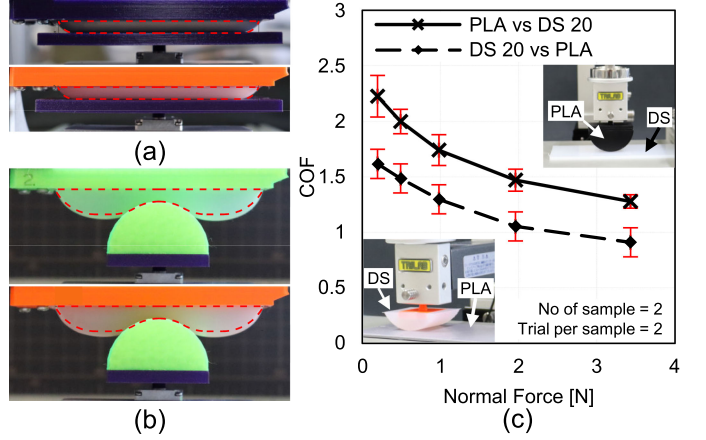


Fig. 5. Theoretical (dotted red curve) and experimental deformed profile of the actuator for 5 mm offset. (a) Full contact and (b) cylindrical contact at 15 kPa. (c) COF measurement test data for DS 20 versus PLA. Experiments were performed for interactions of hard specimen on soft surface (PLA versus DS 20) and for soft specimen on hard surface (DS 20 versus PLA).

Calibrator App”. The nodal coordinates of the actuator in the deformed state were extracted using ImageJ software.¹ The force was measured using a 3-axis force sensor (USL06-H5 Force Sensor, Tech Gihan). The force measurement setup for the actuator in contact with the cylindrical substrate is shown in Fig. 3(e) (90° anticlockwise rotated view) and 5(b). Grasping tests were performed to grasp cylindrical objects using parallel two-fingered shell grippers. The objective of grasping tests is to validate theoretically predicted minimum air pressure and vertical offset h required to attain a grasp.

The theoretical model is based on incompressible neo-Hookean material and requires the material shear modulus as an input parameter. For the experimental tests, we fabricated shell grippers using Dragon Skin (DS) 20 silicone rubber (Smooth-On, Inc., PA). The material properties of DS 20 were evaluated by performing tension tests [29]. The experimental shear modulus was 136 kPa; however, in the simulations, it was assumed to be 175 kPa. The shear modulus was tuned to best fit the theoretical free-space deformation with the experimental deformation. The frictional properties of actuator material are needed for the grasping state simulations. Coefficient of friction (COF) measuring experiments were conducted on a friction measuring machine (TL201 Tt, Trinity-Lab. Inc., Japan). The

¹[Online]. Available: <https://imagej.nih.gov/ij/>

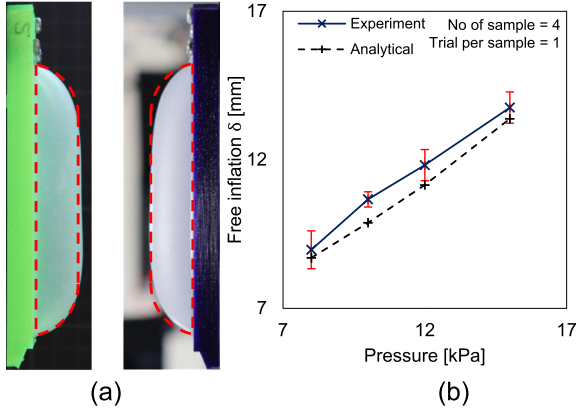


Fig. 6. Free-space deformation of the actuator. (a) Experimental snapshots at 15 kPa and (b) plot of mid-plane extension δ versus air pressure. Dotted red curve in (a) represents theoretical deformed profile of the membrane at 15 kPa.

test setup is shown in the inset of Fig. 5(c). The COF was measured for the contact between the DS and PLA materials and the test data is presented in Fig. 5(c).

V. RESULTS AND DISCUSSION

A. Free-Space Deformation

The snapshots of free-space deformation of the actuators are shown in Fig. 6(a). The theoretical and experimental mid-plane inflation δ of the membrane is presented in Fig. 6(b). The root mean square error (RMSE) for the mid-plane extension is 0.56 mm. The experimental deformed profiles differ slightly from the theoretical profiles. This happens due to boundary conditions of the membrane. In the analytical model, the membrane was assumed to be rigidly clamped at the ends; however, in experiments, the boundaries have little flexibility. In addition, due to manual fabrication, the experimental deformed profiles vary slightly among different actuators [see standard deviations in Fig. 6(b)].

B. Full Contact With Flat Substrate

The theoretical and experimental deformed profiles and contact forces for the actuators in contact with a rigid flat substrate are presented in Fig. 5(a) and 7, respectively. The experimental force was measured for 5 mm and 10 mm offsets. The theoretically estimated contact width [see Fig. 5(a)] matches well with the experimental contact width. The force behavior predicted by the analytical model is also similar to the experimental behavior. The RMSE's for 5 mm and 10 mm offsets are 1.32 N and 2.44 N, respectively.

C. Contact With Cylindrical Object

The theoretical and experimental deformed profiles and contact forces for the actuators in contact with a cylindrical substrate are presented in Fig. 5(b) and 8, respectively. The experimental force was measured for four offsets, i.e., 3, 5, 8, and 10 mm. The contact deformation of the actuator [see Fig. 5(b)] is slightly

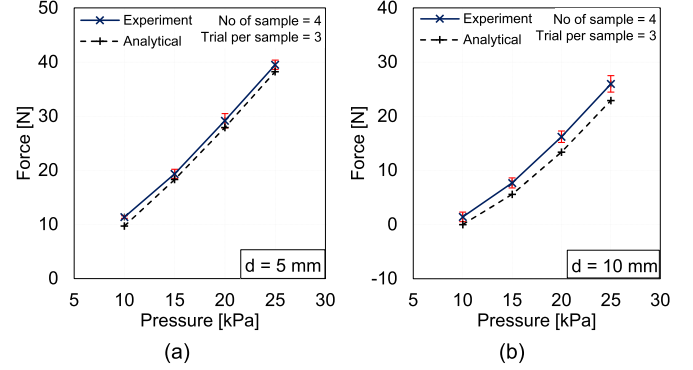


Fig. 7. Theoretical and experimental force plots for actuator in contact with rigid flat substrate. (a) 5 mm offset and (b) 10 mm offset.

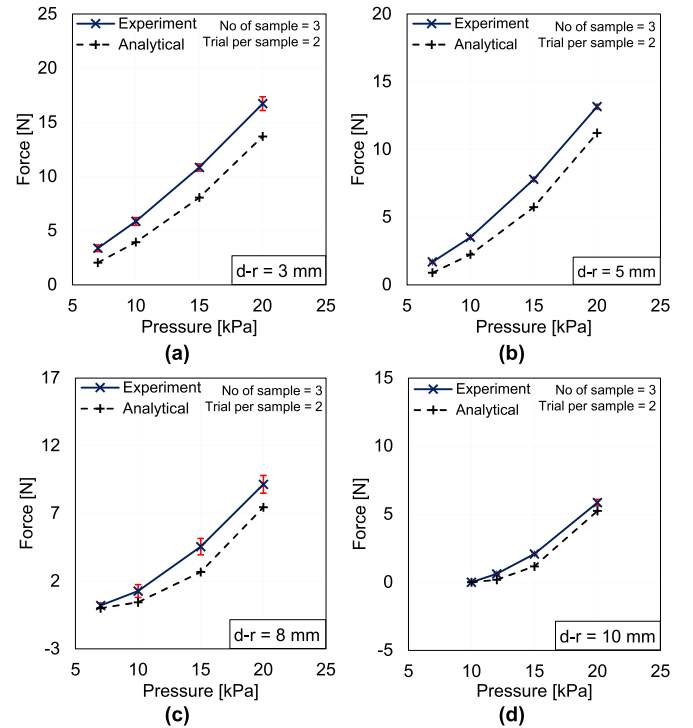


Fig. 8. Theoretical and experimental force plots for actuator in contact with cylindrical substrate at different offsets $d - r$.

smaller than the actual deformation. The predicted force behavior by the analytical model also varies slightly among the offsets. This occurs due to boundary conditions of the membrane and resultant direction of normal contact force. The boundary gets stretched a little with deformation of the membrane and this results in contact area increment and thus additional force. At same magnitude air pressure, the membrane deforms more for lower offset. At 20 kPa, the discrepancy for 3 mm offset is the highest. In experiments, we measured the force generated by the actuator along the Z-direction [see Fig. 3(e)]; however, in simulations, we predicted normal force at the contact region. The contact region exhibits a curved shape, so we calculated the force in Z-direction by projecting the contact area to a plane normal to Z-direction (29). For grasping, we used the normal force directly

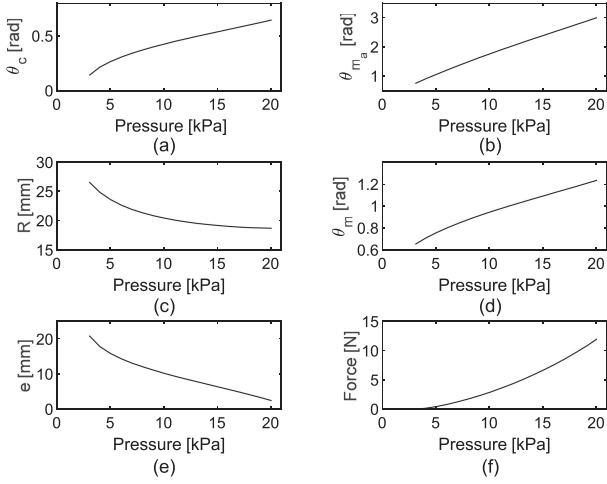
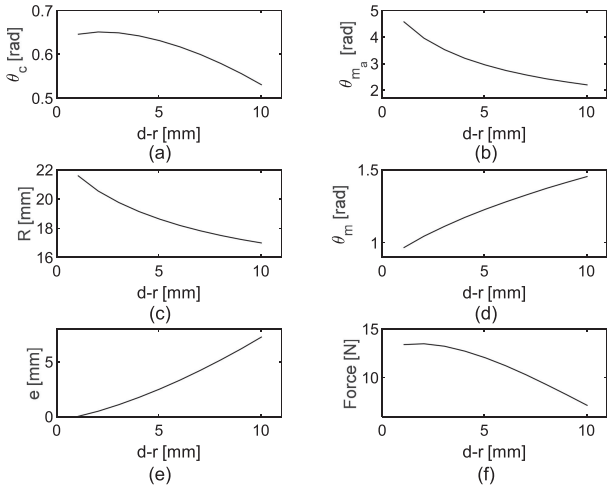


Fig. 9. Parameter variation with pressure for 5 mm offset cylindrical contact.

Fig. 10. Parameter variation with offset $d-r$ for cylindrical contact at 20 kPa air pressure.

to predict the lifting force. The RMSE's for the above-mentioned offsets are 1.33, 1.59, 2.36, and 0.63 N, respectively.

For controlling a parallel gripper, the only adjustable parameters are air pressure and horizontal offset $d-r$. The lifting force required to grasp an object can be achieved by different combinations of pressure and offset $d-r$. In order to understand the deformation behavior of the actuator with respect to pressure and offset $d-r$, we investigated the variation of output parameters involved in the nonlinear algebraic equations via simulations. The variation with air pressure is presented in Fig. 9. Here, the contact angle θ_c represents the contact area and it increases with the increase in pressure. The parameter e represents the portion of membrane with zero curvature and it decreases with the increase in pressure. Practically, this variation seems appropriate. The variation with object offset $d-r$ is presented in Fig. 10. The contact angle in this case decreases with the increase in $d-r$. The portion of membrane with zero curvature increases with the increase in $d-r$. This variation also coincides with experience.

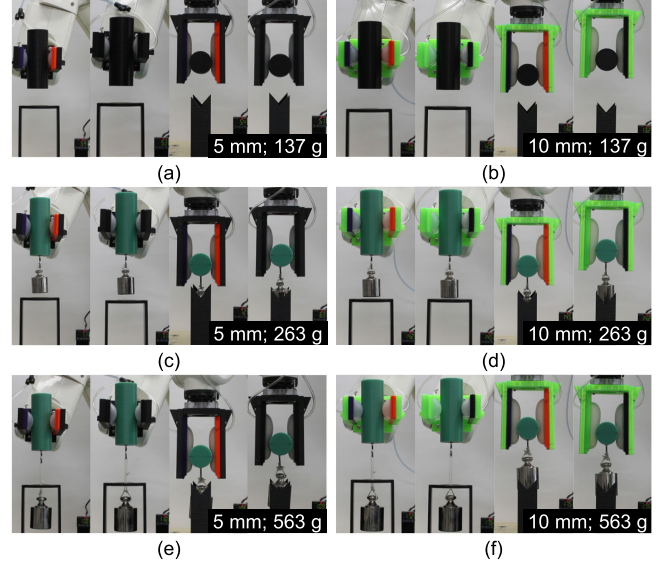


Fig. 11. Experimental photo snapshots of two parallel shell grippers grasping circular cylinders in horizontal and vertical postures.

TABLE II
ANALYTICAL MODEL PREDICTED GRASPING TEST DATA AND EXPERIMENTAL TEST RESULTS

Cylinder		Offset ($d-r$) 5 mm				Offset ($d-r$) 10 mm			
Diameter	Weight	h	p	Grasp Status		h	p	Grasp Status	
(mm)	(g)	(mm)	(kPa)	VP	HP	(mm)	(kPa)	VP	HP
40	137	28.1	5.5	G	G	26.4	13.0	G	G
40	263	30.3	7.2	G	G	27.9	14.5	G	G
40	563	33.7	10.9	G	G	32.1	17.5	G	G

¹ VP – vertical posture of cylinder; HP – horizontal posture of cylinder; G – grasp.

D. Grasping

The COF at the contact region is needed to calculate the normal force required to hold the objects. To determine COF, friction tests were performed with two different setups, having contact conditions similar to the grasping experiments, as shown in Fig. 5(c). For worst case scenario, the test data for DS 20 versus PLA was used in the simulations to determine the normal force. The experimental COF varies with normal force; therefore, the COF taken in the simulations was different for each object. The COF was decided on the basis of required lifting force. We defined lifting force F_L as the force required to hold the object and is applied tangentially to the contact surface (for static case, the lifting force is assumed to be equal to gravity force). The required normal force F_{ext} is then calculated from the lifting force F_L using Coulomb's law of friction as

$$F_{ext} = \frac{1}{\mu_f} \frac{F_L}{2} = \frac{1}{\mu_f} \frac{mg}{2}. \quad (35)$$

Here, we are using a parallel two-fingered shell gripper, so we divided the lifting force per finger. Based on the required normal force, air pressure and vertical offset h were calculated using the analytical model for 5 and 10 mm object offsets. The predicted test data is presented in Table II.

The grasping tests were performed with three 3D printed circular cylinders (PLA material) and two parallel grippers, as shown in Fig. 11. The experiments were performed for both the horizontal and vertical postures of the cylinders. In experiments, the objects were first grasped at the predicted vertical offset, and pressure slightly higher than the predicted value and then, the pressure was reduced to see if the object slips or not at the predicted value. The grippers were able to grasp the objects at predicted pressure and offset.

VI. CONCLUSION

This paper presented an analytical model to calculate the deformation and force characteristics of a flat membrane-based soft actuator. The proposed model predicts the force generated by the actuator for both full and cylindrical contact scenarios. The force characteristics are important for defining the weight-lifting capabilities of the actuator. In this work, we further explored the dependency of cylindrical object grasp on the vertical offset parameter h . The vertical offset is a crucial parameter for the gripper to pick objects. The model provides a relationship between the vertical offset h and other parameters. The grasping test provides insight into the effect of h on the grasp. The presented model is abstract and does not require a CAD file for simulations. It only requires geometric and material parameter values. It can be used with other optimization techniques to find the best-fit design parameters from a large design space (i.e., initial design). The material properties of elastomers are not fixed and in the proposed model we can easily tune them to best-fit the deformation behavior. In the experiments, we used a two-fingered gripper; however, the same modeling approach can be used for other configurations as well, i.e., modular design. The proposed modeling approach can be further extended to investigate the interaction behavior with other surfaces. As a limitation, the presented model do not consider gripper misalignment and asymmetric object grasp; however, in practice, such grasping conditions also encountered.

The future work will investigate eccentric grasp and the optimization of actuator geometry for force and vertical offset h . The model will be further extended to study the interaction of the actuator with soft objects.

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